SOP 02 | Install, Patch, Then Create Database

Scenario 2: build a fresh Oracle 19c Enterprise Edition server on Windows Server 2025 using a patched Oracle Home before DBCA.

Document control | Version 1.0 | Issued 04 November 2026 | Windows x64 | Single-instance Enterprise Edition

USE THIS SOP ONLY FOR A NEW OR REBUILT SERVER. It is not a procedure for the already-working ORCLCDB. Do not run this SOP against your existing database without a separately approved rebuild plan.

Item	Approved baseline / target
Operating system	Windows Server 2025 Standard Evaluation, build 26100 (lab)
Oracle edition	Oracle Database 19c Enterprise Edition, single instance
Oracle Home	C:\app\oracle\product\19.0.0\dbhome_1
Oracle Base	C:\app\oracle
Database / service	ORCLCDB / ORCLPDB1; listener LISTENER TCP 1521
Before / after	Oracle 19.3.0.0.0 → Oracle 19.33 (Windows Bundle Patch)
Scope	Standalone filesystem deployment; no Grid/RAC/ASM

Decision and stop conditions

- Order: extract base image → software-only OUI → update OPatch → apply Windows bundle → create database with DBCA → datapatch/verify.
- Patch files can be downloaded in advance but cannot be applied before Oracle binaries and inventory are installed.
- The Oracle 19.3 base installer is not certified on Windows Server 2025; software-only setup may be possible, but do not override failed OS prerequisite checks. Stop and obtain a supported installation path if blocked.
- If a preexisting Oracle Home, database service or CDB is found, STOP. Preserve existing databases; do not overwrite any home.

Required files: exact selection

Artifact	What to download	Required?
Oracle DB base image	V982656-01.zip (Oracle 19.3 Windows x64 Database Home). For a new installation only.	Scenario 2 only
Windows Database Bundle Patch 19.33	My Oracle Support patch ID 39898799; select Microsoft Windows x64 (64-bit), Oracle Database 19c, Oct 2026.	YES
OPatch utility	My Oracle Support patch ID 6880880; choose compatible 19c Windows x64 build meeting bundle README minimum.	YES
Oracle Client	V982658-01.zip; separate client installer, NOT required for local database server patching.	NO
OJVM / Data Pump patches	OJVM 39803650 and Data Pump 40057600 are conditional, not part of baseline procedure. Verify installed component and exact Windows applicability/readmes.	Conditional
Grid Infrastructure / Linux RU	Do not download the Linux DB RU 39915792 or Windows Grid/RAC bundle for this standalone database.	NO

Do not assume the actual patch ZIP filename. Example naming patterns are p39898799_190000_MSWIN-x86-64.zip and p6880880_190000_MSWIN-x86-64.zip; use the filename presented by the entitled Oracle download page. The patch ID, product, platform, release and README must all match.

- Download through https://support.oracle.com/ using a valid support CSI; an ordinary free Oracle account generally cannot download patches.
- Keep original ZIPs unchanged in C:\OraclePatches\downloads; extract into a separate staging folder. Never extract a bundle directly into ORACLE_HOME.
- Obtain and read the bundle patch README before using any apply command. The README overrides generic commands in this SOP.

PAUSE / CHECKPOINT | G0 — Download authenticity and applicability

GO when: All required archives are present; Oracle patch page and README identify the Windows x64 DB 19.33 bundle for 19c, patch ID 39898799; OPatch platform and minimum version are confirmed.
STOP when: Do not download from mirror sites, use a Linux RU, or apply any unverified patch.

Step 1 — Validate clean Windows Server and pre-stage files

PowerShell conventions and staging

Use PowerShell opened with “Run as administrator” unless SQL> or RMAN> is explicitly shown. Commands beginning with SQL> belong in SQL*Plus; do not paste them directly into PowerShell. All paths are examples grounded in this Oracle Home.

```
$OH = 'C:\app\oracle\product\19.0.0\dbhome_1'
$env:ORACLE_HOME = $OH            # current session only
$env:ORACLE_SID  = 'ORCLCDB'      # current session only
$env:PATH = "$OH\bin;$OH\OPatch;$env:PATH"
$stage = 'C:\OraclePatches\19.33'
New-Item -ItemType Directory -Force -Path $stage | Out-Null
Get-CimInstance Win32_OperatingSystem |
  Select-Object Caption,Version,BuildNumber,OSArchitecture
Test-Path "$OH\OPatch\opatch.bat"
```

Use the downloaded ZIP location appropriate to the machine. Confirm file hashes against My Oracle Support details when provided, not against an unrelated ZIP with a different package layout.

```
Get-FileHash 'C:\OraclePatches\downloads\<actual-patch-file>.zip' -Algorithm SHA256
```

On a fresh host, OPatch inventory checks are NOT run yet: the Oracle Home does not exist until you complete the software-only installer. After Step 3, verify opatch version and lsinventory before replacing the utility.

```
Get-CimInstance Win32_Service | Where-Object Name -match 'Oracle' |
  Select-Object Name,State
Test-Path 'C:\app\oracle\product\19.0.0\dbhome_1'
Test-Path 'C:\OraclePatches\downloads\V982656-01.zip'
Get-Volume | Select-Object DriveLetter,FileSystemLabel,SizeRemaining
```

PAUSE / CHECKPOINT | S2-01 — Clean host and inputs

GO when: OS is Windows Server 2025 x64, enough RAM/disk are available, V982656-01.zip and 39898799/6880880 downloads are ready, Oracle Home is empty/new and no existing DB is at risk.
STOP when: If the Oracle Home is populated or Oracle services already exist, do not overwrite them.

Step 2 — Extract 19.3 database image to Oracle Home

Oracle 19c for Windows is image-based: extract the database ZIP into the final home, not a transient Downloads folder. Do not use V982658-01 (Oracle Client).

```
$baseZip = 'C:\OraclePatches\downloads\V982656-01.zip'
$OH = 'C:\app\oracle\product\19.0.0\dbhome_1'
New-Item -ItemType Directory -Force -Path $OH | Out-Null
Expand-Archive -LiteralPath $baseZip -DestinationPath $OH
Test-Path "$OH\setup.exe"
Get-ChildItem $OH | Select-Object -First 12 Name
```

PAUSE / CHECKPOINT | S2-02 — Image extracted

GO when: setup.exe is directly under the intended Oracle Home and bin, oui, rdbms subdirectories are present; no nested extra top-level folder.
STOP when: Do not launch setup from Downloads or a wrongly nested directory.

Step 3 — Run Oracle Universal Installer (software only)

Right-click C:\app\oracle\product\19.0.0\dbhome_1\setup.exe and select Run as administrator. Use these exact OUI choices:

Wizard screen	Selection / value
Select Configuration Option	Set Up Software Only
Database Installation Options	Single instance database installation
Database Edition	Enterprise Edition
Oracle Home User	Use Virtual Account
Oracle Base	C:\app\oracle
Software Location	C:\app\oracle\product\19.0.0\dbhome_1
Prerequisite Checks	All mandatory checks pass without overrides
Summary	Review every setting, then Install
Finish	Oracle software registration completed successfully

PAUSE / CHECKPOINT | S2-03 — Software install complete

GO when: OUI Finish reports success, SQL*Plus and DBCA files exist, Oracle Home and Oracle Base registry entries are correct, no CDB/PDB has been created.
STOP when: Do not bypass operating-system or other failed prerequisites. Do not start DBCA yet.

```
Test-Path "$OH\bin\sqlplus.exe"
Test-Path "$OH\bin\dbca.bat"
& "$OH\bin\sqlplus.exe" -v
& "$OH\OPatch\opatch.bat" version
& "$OH\OPatch\opatch.bat" lsinventory
Get-ItemProperty 'HKLM:\SOFTWARE\ORACLE\KEY_OraDB19Home1' |
  Select-Object ORACLE_HOME,ORACLE_BASE,ORACLE_HOME_NAME
```

Step 4 — Update OPatch in the software-only Home

Update the OPatch utility

1. Extract the downloaded OPatch ZIP into C:\OraclePatches\opatch_extract (NOT the database patch folder). Verify that it contains a top-level OPatch directory.

```
$opatchZip = 'C:\OraclePatches\downloads\<actual-6880880-zip>.zip'
$extract = 'C:\OraclePatches\opatch_extract'
Expand-Archive -LiteralPath $opatchZip -DestinationPath $extract -Force
Test-Path "$extract\OPatch\opatch.bat"
```

2. Back up the old OPatch folder; replace it with the extracted OPatch directory. Avoid merging folders, as stale files could remain. Ensure no Oracle patch utility is running.

```
Rename-Item -LiteralPath "$OH\OPatch" -NewName 'OPatch_before_1933'
Copy-Item -LiteralPath "$extract\OPatch" -Destination "$OH\OPatch" -Recurse
& "$OH\OPatch\opatch.bat" version
```

3. Compare OPatch version with the minimum version in the 19.33 Windows bundle README. A higher compatible version is acceptable if the README permits it.

PAUSE / CHECKPOINT | G2 — OPatch readiness

GO when: New OPatch folder contains opatch.bat; “opatch version” works; version meets the 19.33 README; previous OPatch folder remains available for recovery.
STOP when: If opatch version fails, stop; do not attempt an Oracle database bundle patch.

Step 5 — Apply the Windows bundle before DBCA

Unpack the Windows Database Bundle Patch

```
$bundleZip = 'C:\OraclePatches\downloads\<actual-39898799-zip>.zip'
$stage = 'C:\OraclePatches\19.33'
Expand-Archive -LiteralPath $bundleZip -DestinationPath $stage -Force
Get-ChildItem -LiteralPath $stage -Recurse -Filter README* |
  Select-Object FullName
Get-ChildItem -LiteralPath $stage -Recurse -Directory |
  Where-Object Name -eq '39898799' | Select-Object FullName
```

An Oracle patch ZIP commonly extracts a folder named 39898799. Your actual layout may differ. Locate the patch-root directory containing etc, files and inventory and set $PatchDir to that exact location. Do not guess the folder from the ZIP filename.

```
$PatchDir = 'C:\OraclePatches\19.33\39898799' # adjust after inspection
Test-Path "$PatchDir\etc"
Test-Path "$PatchDir\files"
& "$OH\OPatch\opatch.bat" prereq CheckConflictAgainstOHWithDetail -ph $PatchDir
```

PAUSE / CHECKPOINT | G3 — Patch prerequisites

GO when: README has been reviewed; the extracted root is correct; opatch prereq reports that conflict checks are satisfied and no mandatory dependency is missing.
STOP when: Stop if there is a conflict, wrong platform, missing prereq, or failed inventory. Do not use -force to bypass conflicts.

Apply the patch to the Oracle Home

The database services must be stopped for Scenario 1. Scenario 2 has no database instance yet, but any background Oracle process from the new Oracle Home must still be closed.

```
Set-Location -LiteralPath $PatchDir
& "$OH\OPatch\opatch.bat" apply
```

Follow the on-screen prompts and the exact 19.33 Windows README instructions. The generic command shown here is appropriate only when README specifies standard in-place OPatch application. Never terminate the patch process halfway through.

```
& "$OH\OPatch\opatch.bat" lsinventory
& "$OH\OPatch\opatch.bat" lspatches
```

PAUSE / CHECKPOINT | G4 — Binary patch successful

GO when: OPatch ends with success and inventory clearly lists the expected 19.33 Windows Database Bundle Patch 39898799 (or its README-documented constituent patch identifiers).
STOP when: If OPatch fails or inventory lacks the target patch, STOP before running DBCA or datapatch. Inspect the OPatch

log and use Oracle's documented recovery procedure.

Step 6 — Create the CDB and PDB using DBCA

Once binary patching passes, launch DBCA using the graphical Windows program or directly from the Oracle Home.

```
& "$OH\bin\dbca.bat"
```

DBCA screen / field	Recommended configuration
Database Operation	Create a database
Creation Mode	Advanced configuration
Deployment	Oracle Single Instance Database; General Purpose or Transaction Processing
Global database name / SID	ORCLCDB / ORCLCDB
Container Database	Checked; local undo enabled
Number/name of PDBs	1 / ORCLPDB1
Storage	Use template file; filesystem. Confirm datafiles under C:\app\oracle\oradata
Fast Recovery Area	Filesystem, under C:\app\oracle\fast_recovery_area; choose based on capacity
Logging	Enable ARCHIVELOG if RMAN/point-in-time recovery labs are planned
Listener	LISTENER TCP 1521
Database Vault/OLS	Leave unchecked unless licensed/required
Memory	ASMM; size based on server RAM (prior lab used ~4.9 GiB database memory)
Character sets	AL32UTF8 / AL16UTF16; language American, territory India
Connection mode	Dedicated server
Sample schemas	Leave unchecked; install later for admin exercises
EM Express	Optional; port 5500, lab-only restricted access
Administrative passwords	Unique strong SYS, SYSTEM, PDBADMIN; do not embed passwords in logs
Creation options	Create database; optionally generate scripts
Summary	Review all entries and filesystem free space before Finish

PAUSE / CHECKPOINT | S2-04 — DBCA configuration approved

GO when: Correct CDB, PDB, memory, storage, listener, recovery and character set values appear on Summary; passwords are securely stored.
STOP when: If summary differs, use Back to correct values before creating the database.

4. Click Finish and let DBCA complete. Review output; do not accept a partial or failed CDB/PDB creation.

```
$env:ORACLE_SID = 'ORCLCDB'
& "$OH\bin\sqlplus.exe" / as sysdba
```

```
SELECT name,open_mode,log_mode,cdb FROM v$database;
SHOW PDBS;
SELECT con_id,name,open_mode FROM v$containers ORDER BY con_id;
EXIT;
```

PAUSE / CHECKPOINT | S2-05 — Database created and open

GO when: DBCA reports success; ORCLCDB is READ WRITE and CDB=YES; PDB$SEED is READ ONLY; ORCLPDB1 is READ WRITE; listener service is ready.
STOP when: If DBCA fails, review DBCA logs and do not proceed to application/schema creation.

Step 7 — Run datapatch for a newly created database

Although the Oracle binaries were patched before DBCA, ensure the newly created database SQL components are at the intended release level. Oracle documentation explicitly includes a datapatch step after database creation. Do not presume binary patching is the final step.

```
Get-ChildItem "$OH\OPatch\datapatch*" | Select-Object Name
& "$OH\OPatch\datapatch.bat" -verbose
& "$OH\bin\sqlplus.exe" / as sysdba
```

```
SELECT con_id,patch_id,action,status,description
FROM cdb_registry_sqlpatch
ORDER BY action_time,con_id;
SHOW PDBS;
EXIT;
```

PAUSE / CHECKPOINT | S2-06 — New database SQL patch level aligned

GO when: datapatch executed successfully (or README explicitly documents a no-op); root/seed/ORCLPDB1 patch state matches the 19.33 Oracle Home inventory.
STOP when: If registry does not align with the binary patch, stop and investigate datapatch logs; do not declare installation complete.

Step 8 — Startup configuration and final acceptance

Save PDB state to ensure it reopens when the CDB starts; confirm Windows service autostart. These are distinct from having an Oracle database installed.

```
& "$OH\bin\sqlplus.exe" / as sysdba
```

```
ALTER SESSION SET CONTAINER=CDB$ROOT;
ALTER PLUGGABLE DATABASE ORCLPDB1 SAVE STATE;
SELECT con_name,state,restricted FROM dba_pdb_saved_states;
EXIT;
```

Final acceptance and service tests

```
& "$OH\OPatch\opatch.bat" lsinventory
Get-CimInstance Win32_Service |
  Where-Object Name -match 'Oracle' |
  Select-Object Name,State,StartMode
& "$OH\bin\lsnrctl.exe" status LISTENER
Test-NetConnection 127.0.0.1 -Port 1521
```

```
& "$OH\bin\sqlplus.exe" / as sysdba
```

```
SELECT instance_name,version_full,status FROM v$instance;
SELECT name,open_mode,log_mode,cdb FROM v$database;
SHOW PDBS;
SELECT con_id,patch_id,action,status
FROM cdb_registry_sqlpatch
ORDER BY action_time DESC,con_id;
```

Test a real service connection, not only TNSPING. SQL*Plus password prompts may need quoted input when a password contains @; do not place passwords in command lines, scripts or runbook evidence.

```
& "$OH\bin\sqlplus.exe" /nolog
```

```
CONNECT SYSTEM@"127.0.0.1:1521/orclpdb1"
-- Enter administrative password at prompt; if it contains @,
-- enclosing password in double quotes worked in this lab.
SHOW CON_NAME;
SHOW USER;
```

PAUSE / CHECKPOINT | G6 — Release acceptance

GO when: OPatch inventory reports the intended bundle, SQL registry shows SUCCESS in required containers, ORCLPDB1 is READ WRITE, Oracle services are running, listener responds on 1521 and authenticated SQL*Plus connects.
STOP when: If any acceptance test fails, do not sign off the change. Record full error and relevant OPatch/sqlpatch/listener logs.

Install / rebuild evidence

Archive: OS/build checks, all ZIP filenames and hashes, patch README, installer summary, OPatch version/inventory, conflict check, OPatch success logs, DBCA completion summary, datapatch output, SQL patch registry, listener and user connection tests.

Source references and operator notes

Oracle: Windows OS minimum version (19.27 for Server 2025): https://docs.oracle.com/en/database/oracle/oracle-database/19/ntdbi/operating-system-checklist-oracle-database-installation-microsoft-windows.html

Oracle Database 19c Windows downloads: https://www.oracle.com/database/technologies/oracle19c-windows-downloads.html

Oracle Database Patch Maintenance 19c: https://docs.oracle.com/en/database/oracle/oracle-database/19/dbptc/index.html

Oracle: database creation and datapatch: https://docs.oracle.com/en/database/oracle/oracle-database/19/admin/creating-and-configuring-an-oracle-database.html

My Oracle Support - authenticated patch downloads and READMEs: https://support.oracle.com/

Oracle Windows DB 19.33 patch ID cross-check (third-party tracker, NOT a substitute for MOS): https://www.markusdba.de/2026/11/01/oracle-database-19c-auf-windows-ru-patch-nummern/

Authority rule: My Oracle Support patch record and its README govern platform, prerequisites, specific operations and rollback. This runbook provides a controlled sequence, not a replacement for Oracle's patch-specific instructions.